

Early Childhood Intervention System

Submitted in partial fulfillment of the requirements for the degree of

Master of Science in Data Science

by

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DECLARATION

I hereby declare that the Dissertation -I titled "**Early Childhood Intervention System** " submitted by me, for the award of the degree of *Master of Science in Data Science* to VIT is a record of bonafide work carried out by me under the supervision of **Dr. Sandip Dalui**.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place : Chennai

Date : 19/11/2025

Signature of the Student

Pallavi K – 24MDT1045
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CERTIFICATE

This is to certify that the Dissertation -I titled “**Early Childhood Intervention System**” submitted by **Pallavi K (Reg. No.: 24MDT1045), Department of Mathematics, School of Advanced Sciences, VIT Chennai**, for the award of the degree of *Master of Science in Data Science*, is a record of bonafide work carried out by her under my supervision during the period, **09.07.2025 to 14.11.2025**, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The thesis fulfills the requirements and regulations of the University and in my opinion meets the necessary standards for submission.

Place : Chennai

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ABSTRACT

The initial years of life are marked by a very long sequence of substantial changes that characterize the whole period of childhood—especially the first ones, which are greatly important for the nature of children's thoughts, feelings, and connections with the world. In fact, it is extremely different to grow up in the present time than in the case of the kids' parents. Throughout the first eight years, the brain development of children becomes most extensive that very fast. Besides, they have to acquire the skills to talk, express their emotions, and perform the simplest acts for getting along during this period. The problem with omnipresent screens is that there is too much unfiltered screen time which alters the language-learning ability of the kids, their attention span, sleep pattern, playing with friends, and even how they manage calming down. It is the parents that play a significant role in being part of the solution. They not only limit children's views and actions but also the time spent on different activities. Next, we have the late childhood period from 8 to 12 years which is equally important but for some reason not as much attention is given to it. The kids start to have a better understanding of themselves and become more dependent on their peers; they assert their independence and use technology for playing games, for school apps, and for social media. Then, of course, there is the freedom children have to use screens. Each of those has its own difficulties: the distraction, the mood swings, and the growing number of unhealthy screen habits that may accompany all of that. There is a lot of research work done on very young children, but there is not so much about what happens when they grow older, how their media habits start to transform, what keeps things under control. The present thesis is presenting that question. Based on research in psychology, digital media, and technology studies, the emerging challenge is how to measure screen time in early and later childhood, how to recognize screen time problems when they arise, and how to learn. Various contexts have been incorporated into the scenario to provide improved services for children and families. Therefore, the ideal way seems to be to supervise the children and to offer them intuitive dashboards and AI-powered smart gadgets that will provide individual assistance. However, there are not enough efficient tools for parents to keep an eye on their children's digital activities during the preschool to preteen stages. It is the time when families urgently require genuine solutions. And the system I am detailing will enable parents and families to direct their children on how to have a healthy and balanced approach to screen time throughout their lives, considering every age and stage.

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List of Abbreviations

Abbreviation	Definition
AAP	American Academy of Pediatrics
API	Application Programming Interface
CSV	Comma-Separated Values
EDA	Exploratory Data Analysis
F1-score	F1-Score (a measure of a test's accuracy)
GUI	Graphical User Interface
MAE	Mean Absolute Error
MSE	Mean Squared Error
RMSE	Root Mean Squared Error

1.INTRODUCTION

Presently, children are not just playing with apps on their mobile phones but also making the app count down days to their birthday. The use of mobile devices in the family has become so common that it is almost like a daily interaction everywhere. The usage of these gadgets and apps by a large number of children has greatly impacted their digital habits. Children allow digital apps to invade their lives and at the same time, they make them the central part of their activities. This entire scenario has made researchers, educators, and parents more alert, each of them being eager to know the development, learning, and emotional aspect of the children in this situation.

1.1. OBJECTIVE

The primary goal of the research with the investigation of `app_usage_dataset_10k.csv` file, which presents the app usage of little kids from various age groups over 10,000 sessions, is to figure out how kids have a strong preference for particular apps. Our goals are: 1. To gain insight into children's interactions with app categories (Games, Education, Social, Utility) and their emotions' influence on the kids' app preferences. 2. To build a forecasting model of the optimal timing for adult intervention that is based on the app usage duration, frequency, and the time of the day. 3. To investigate the three age groups 03-05, 06-08, 09-12 in terms of differences in the usage patterns. 4. To identify the trend that can assist parents, educators, and app developers to nurture good digital habits.

1.2. MOTIVATION

Why is this so? A three-year-old child can already be diagnosed with a screen addiction and it is crucial to have an idea of both the time and the different activities of the child. The statistics suggest that kids are using apps more than once over a period of time (i.e., an hour or more) which shows the necessity to categorize such uses into healthy and unhealthy ones. The dataset does not merely keep time but also registers the mood of the children — happy, neutral or sad — later on. This is a powerful symptom that the apps are affecting their moods, and maybe even their mental health, and the children are thereby indirectly led by the apps to experience moods, and mental health, and thus indirectly be led by the apps to experience moods and mental health. Besides that, almost 1 in 3 app sessions signals the need for the involvement of an adult. The point is clear, we need a more competent strategy to identify the moment when

things start going wrong. Therefore, all this has to been leading us to look for genuine ways to support parents and teachers in directing children's digital lives along the right path. By spotting the trends, we will be in a position to develop tools that will alert adults before a problem becomes serious.

1.3. BACKGROUND

(i) Patterns and Determinants of Screen Use : It is a combination of different things: the devices that are accessible to them, the content that they find appealing, the way their parents behave, their daily routines, and also the presence of their siblings or friends. Currently, the main screens for preschoolers are tablets and smartphones; the TV is no more (Kabali et al., 2015; Rideout et al., 2020). Children's activities on these devices are mostly limited to playing games, video watching, and educational apps, and occasionally, they are unwittingly getting exposed to inappropriate content. The parents' influence is of great importance. Parents who apply rules, take part in co-viewing and help children to adopt good habits have managed to reduce the screen time of their kids (Jago et al., 2014; Carson et al., 2015). On the other hand, parents who are busy all the time or limit children's activities usually provide more screen time for their children (Tandon et al., 2012; Wu et al., 2022). The influence of siblings and friends is huge, however, unlike other factors, we know very little about it.

ii) Prolonged exposure to the screen presents harm on the emotions and well-being of the child: Prolonged use of apps can cause shorter attention span, less concentration, and even slow development of reading skills among children (Hutton et al., 2019; Pagani et al., 2010). When more attention is given to screens than to each other, the parents and kids communication is lessened, this could result in a child's language impairment (Christakis, 2009; Wu et al., 2017). Additionally, children may display more behavioral problems in school, such as inability to focus, having tantrums, or getting into fights with others. On the negative side, sleep is disrupted, kids are less active, and consequently, obesity may become an issue. However, through monitoring and directing children's use of the internet by parents, child development may thus be enhanced rather than hindered. So, to sum it up, it is not just a matter of how long kids are glued to the screens, but also a matter of what they are doing and with whom they are doing it.

iii) Parental Involvement and Behavioral Intervention: Parents have always been the basic pillars of protection. Programs that not only teach parents how to impose restrictions but also provide them with concise guidelines, significantly affect the reduction of screen time and

consequently, the enhancement of children's behavior. Some resources, like the MINISTOP application, facilitate the combination of screen time management with daily family routines, physical activity, and overall better lifestyle (Delisle Nyström et al., 2015; Poonia et al., 2024; Yilmaz et al., 2015).

iv). Measurement and Assessment Tools: Measuring the exact amount of screen time per day is quite difficult. The majority of parents complete the surveys, nevertheless, let's face it—most of them tend to miscalculate the numbers (Jago et al., 2014). The PMUM tool can recognize some of the earliest signs of the problem, for instance, when children can't detach themselves from their gadgets and begin to miss out on their daily routines (Domoff et al., 2019). It would be perfect to rely on direct observation and long-term studies, however, they are quite demanding in terms of time and money. They simply do not work on a larger scale. Recently, digital monitoring tools are increasingly becoming the choice for the tracking of children's activities on their devices. These monitoring systems are so advanced that they can tell the exact applications that the kids use, the duration of the usage and even the type of content that is viewed—all these are being done in real time. The monitoring is so high tech that it gives us hard data rather than making guesses. It does bring out very clearly how kids make use of their gadgets and it is even instrumental in coming up with solutions that fit the real situation and not just based on assumptions.

v). Cross-Cultural Perspectives and Policy Guidelines: All kids, no matter where they are located, use screens a lot, but culture is the main factor that defines the rules that parents follow—like whether they are there watching along with their kids or how strict they are concerning device usage. Families in North America and Europe have some similarities and differences in their way of life, but when it comes to screen time each country has its own approach (Tremblay et al., 2017; Storz et al., 2025). Moreover, factors such as family income and the nature of the available home environment are very influential in determining the kind of devices that kids will have access to and the content that they will view.

2.PROJECT DESCRIPTION

Through the use of actual data, this project investigates the way children mobile apps and not merely the discussions around “too much screen time.” We used 10000 detailed dataset for the model training. What remarkable attribute is contained in this great data set? From all aspects, it is the most complete app usage data; with it, It will not only be the case to be precise about the duration and frequency of app usage by kids but also the kinds of feelings (happy, sad, or neutral) they experienced and the moments when possibly an adult's help would be needed. The preserving of the patterns will be done by age groups (3-5, 6-8, 9-12) and by application categories (Games, Education, Social, Utility). The last word in the project will be a machine learning model that will identify the dangerous part of a child's app session; thus, it will be known when parents or teachers should interfere. The model keeps track of negative emotions that are often expressed, long sessions, and late-night usage. We are trying to give parents and educators something truly useful by turning their attention to the real issue.

PROJECT GOALS

- i. Monitor the change of children's app behavior over time.
- ii. Detect warning signs for interventions that are the largest.
- iii. Create a model that can forecast risky usage.
- iv. Examination of the app usage of children and their emotions.
- v. Analyze the interrelations of categories of applications, in usage accordingly to usage trends, special periods that require children's guidance.
- vi. Study the impact of different apps on the overall health of children.

3. TECHNICAL SPECIFICATIONS

We trained the model in python. Our road map is as follows:

- The first step in data analysis will be the use of Pandas to read, cleanse and manipulate app_usage_dataset_10k.csv.
- Since Matplotlib and Seaborn are visual data interpretation tools, they will thus make our data understanding much easier by showing us the trends in usage, feelings, and intervention needs.
- The data will be split using scikit-learn, and a Random Forest Classifier will be trained to predict the time for intervention, and the accuracy of the model will be verified.

- The model will be saved using Joblib after training, so it can be accessed for future use and predictions without going through the retraining process.

4.DESIGN APPROACH AND DETAILS

MATERIALS, APPROACH AND METHODS

The project will sequentially apply a Python-centric data science stack for the processing of `app_usage_dataset_10k.csv` file and the development of a predictive intervention system. The strategy adopted is very systematic, going from comprehending the data and exploratory analysis to predictive modeling, wherein each next step is a direct response to the project goals.

Pandas: The main library that is going to be used in this work is Pandas. Initially, the dataset is read through the CSV file into a DataFrame (the primary data structure that we are utilizing) along with it. The subsequent activity is data manipulation, which involves cleaning and filtering, grouping either by an `age_group` or `app_category`, and calculating summary statistics for `duration_min` and `frequency_last_7d`. This is all done by means of the very user-friendly data handling facilities provided by Pandas.

NumPy: NumPy is going to be employed for fast numerical operations; this will be particularly the case while we are performing computations on arrays and dealing with the numerical outcomes of our machine learning model.

Matplotlib & Seaborn: It is impossible to overlook the significance of these two visualization libraries during the process of exploratory data analysis and the whole way of communicating our findings. Matplotlib will serve as the foundation for simple plots, whereas Seaborn will be the preferred library for performing complicated and more attractive statistical visualizations. The same will be done through histograms' creation for determining the usage duration distribution, count plots for showing the proportions of different emotions and intervention flags, and heatmaps for pinpointing relationships between various factors.

Scikit-learn: Alternately, the core library that will be largely operated within our machine learning system will be Scikit-learn. A particular function that accompanies this library, `sklearn.model_selection.train_test_split`, is the one that we will use for the data

separation into the training and testing sets. Dividing the dataset is the only way you are going to have an accurate idea of the performance of your model, and it prevents you from being deceived by overfitting to the data you have already seen.

sklearn.ensemble.RandomForestClassifier: Furthermore, it is not prone to overfitting as easily as a single Decision Tree. Thus, it is very suitable for the classification task of predicting the `intervention_required` flag - either 0 or 1.

skullmetrics: We depend heavily on the classification report among all the different metrics since it presents not only the precision but also the recall and the F1-score, which combined provide a deeper understanding of our model's ability to detect the positive class, which consequently means intervention time.

Joblib: Once our model is trained and assessed, the next step will be to use the Joblib library to convert the model object into a byte stream and save it to a file named `intervention_model.pkl`, for instance. This way, the model becomes persistent and can be loaded by a different application or script to carry out real-time predictions without the need to retrain, which fulfills the "system" requirement of the project.

Methodological Steps:

1. Data Loading and Profiling - At first, the data will be loaded using Pandas library. The data types will be checked, and missing values will be counted, while descriptive statistics will be provided to give a general idea of the data characteristics, such as the mean usage duration and the distribution of the age groups.

2. EDA: we will perform an Explonatory distributed analysis to analyse patterns and correlations. This includes:

- Usage trend analysis by depicting the distribution of `duration_min` and `frequency_last_7d` columns across `app_category` and `age_group` using bar charts and box plots as visualization methods.
- The first step in our analysis was to look at what emotions led to which outcomes by creating a cross-tabulation and recording the number of emotions for plotting against `app_category`.
- The next step was to determine the reasons for the intervention by comparing the `intervention_required` column with the features `time_of_day` and `app_category`.

3. Feature Engineering and Preprocessing: The categorical attributes `app_category`, `time_of_day`, `emotion` and so on, will be subjected to one-hot encoding which will facilitate their conversion into numbers. Each category will have binary columns created for it and this will make it easier for the Random Forest algorithm to understand the input data correctly. We will designate our feature set as `X` and the dependent variable `y` will be the column of `intervention_required`.

4. Model Building and Training: The function `train_test_split` is utilized to separate the data that has been preprocessed into two portions, with one part meant for training and the other part for testing. First declare a `RandomForestClassifier` model. By this time, the model is already aware of the hidden patterns that connect usage behaviors and the need for intervention, thus it learns the associations that the model is required to capture.

5. Model Evaluation: The model's accuracy will be measured by `accuracy_score` and `classification_report` that will be generated based on the comparison of these predictions and true values. This process is very crucial as it will ascertain the model's dependability and usefulness in the real-world scenario.

4.2 CODES AND STANDARDS

- Evaluating Model Performance

```
Logistic Regression Model Evaluation:
Accuracy: 0.89
Precision: 1.00
Recall: 0.63
F1-Score: 0.77
ROC AUC: 0.95

Classification Report (Logistic Regression):
              precision    recall  f1-score   support

     0               0.86       1.00       0.92       1391
     1               1.00       0.63       0.77        609

   accuracy               0.89       2000
  macro avg               0.93       0.81       0.85       2000
 weighted avg               0.90       0.89       0.88       2000
```

```
Random Forest Classifier Model Evaluation:
Accuracy: 0.88
Precision: 0.85
Recall: 0.73
F1-Score: 0.79
ROC AUC: 0.95
...
      accuracy          0.88      2000
      macro avg      0.87      0.84      0.85      2000
      weighted avg    0.88      0.88      0.88      2000
```

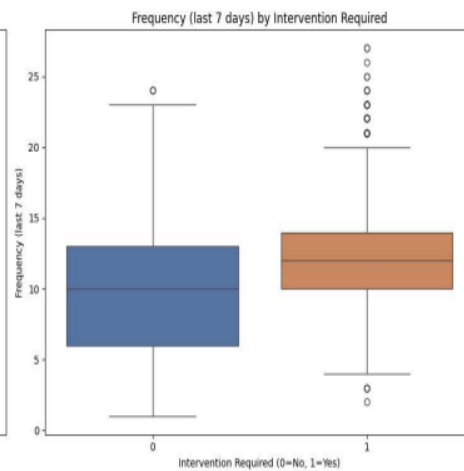
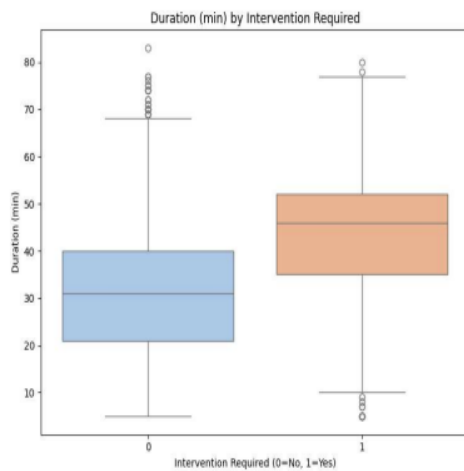
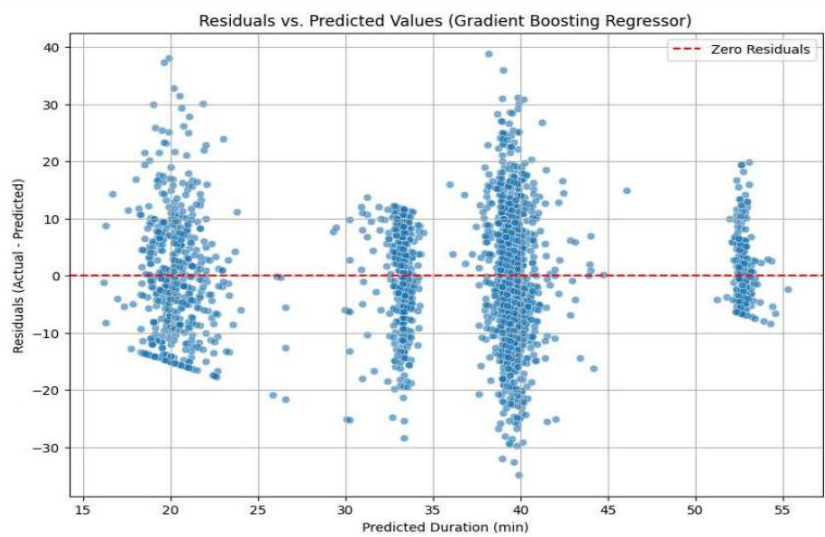
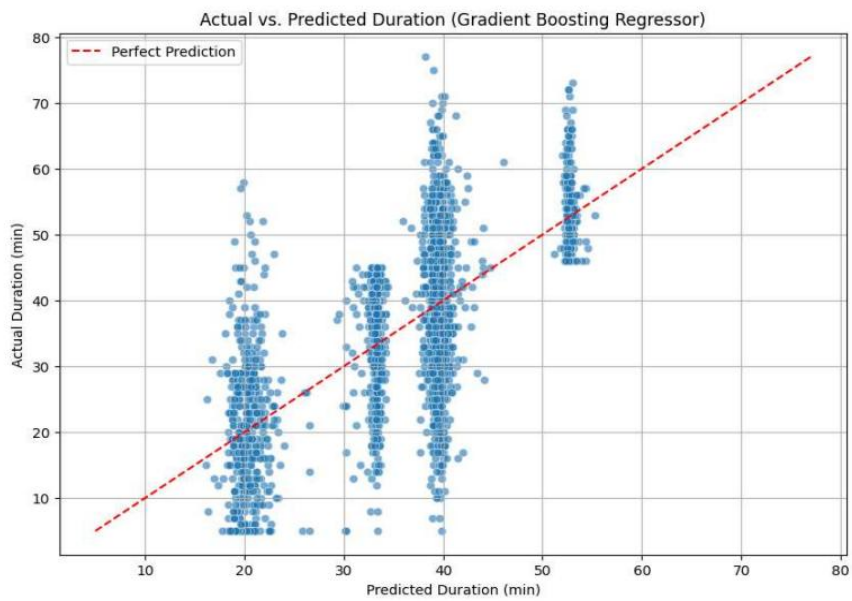
- Saving the Trained Model for Deployment

```
Model Performance Metrics DataFrame:
      Model      MAE      MSE      R2
0  Linear Regression  8.749004  120.508144  0.432970
1    Decision Tree  11.920083  233.927347 -0.100705
2    Random Forest   9.337350  138.958116  0.346156
3  Gradient Boosting   8.463140  113.695413  0.465026
```

5.SCHEDULE, TASKS AND MILESTONES

S.NO.	MONTH-WEEK	PLAN
1.	JULY - WEEK 2	Identification and finalization of the research problem, objectives, and scope of the Early Childhood Intervention System.
2.	JULY - WEEK 3	Comprehensive literature review on children's digital behavior, the impact of screen time, and existing predictive intervention models.
3.	JULY - WEEK 4	Setup of the technical environment (Anaconda, Jupyter) and initial loading and profiling of the <code>app_usage_dataset_10k.csv</code> .
4.	AUGUST - WEEK 1, 2	Perform Exploratory Data Analysis (EDA) to visualize usage patterns, emotional responses, and intervention distributions across age groups and app categories.
5.	AUGUST - WEEK 3, 4	Conduct in-depth analysis to find correlations between app usage metrics (duration, frequency, time of day) and the <code>intervention_required</code> flag.
6.	SEPTEMBER - WEEK 1, 2	Feature engineering and data preprocessing (e.g., one-hot encoding) to prepare the dataset for machine learning model training.
7.	SEPTEMBER - WEEK 3	Build and train the <code>RandomForestClassifier</code> model to predict intervention requirements based on the engineered features.
8.	SEPTEMBER - WEEK 4	Evaluate model performance using key metrics (accuracy, precision, recall, F1-score) and refine the model if necessary.
9.	OCTOBER - WEEK 1, 2	Interpretation of final results, creation of key project visualizations, and saving the trained model using Joblib for future deployment.
10.	OCTOBER - WEEK 3, 4	Drafting the final project report, including all sections from the introduction to the conclusion, and documenting the code and methodology.
11.	NOVEMBER - WEEK 1,2	Final review of the project and report, incorporating feedback from the guide, and preparation for final submission.

6.PROJECT OUTPUTS



7.RESULTS AND DISCUSSION

Higher inspection would be given on the app_usage_dataset_10k.csv file and its consequences for the digital wellbeing of their tiny tots. The important discoveries will be introduced one at a time, starting from children's app usage and finally reaching the accuracy of our intervention prediction model.

App Usage Pattern

To begin with, we focused on the main aspects, meaning we looked at who was using which apps and how long the usage was. The data showed that most children did not spend much time on the apps, however, there were some really long sessions. The distribution was such that quite a few kids spent over an hour continuously using an app.

As we looked at the usage patterns day by day, the time did come up with the pattern. Afternoon and evening are opportune periods to access applications. which matched with kids returning home from school. However, actually, the usage just before bed being so heavy really made us question if it might be a problem. That indeed is something that can disturb sleep.

The most used and the longest time-taking apps were no surprises; games were the clear winners in both categories. The next in line were social apps and then educational ones. The usage of utility apps was almost negligible which is quite rational as kids might only access them for quick, specific tasks. There was also variation according to the age group. Older children aged 9 to 12 years were found to use social applications for a considerably longer period when compared to younger children aged 3 to 5 years, whereas the latter group barely spent more time on educational apps.

Concordant Patterns in Use and Emotional Functions Leading to Escalations of Interventions

We further tried to see how app use was related to the children's emotions described by them. It is quite exciting at this point: the game apps were leading in "happy" responses, thereby indicating that they are still very much enjoyed by the kids. Yet, there came a lot of "sad" reactions for the games as well—most likely due to frustration or losing. On the other hand,

social media apps exhibited a higher number of "sad" and "neutral" feelings, which could be associated with issues such as social comparison, cyberbullying or just a case of passive scrolling.

This is basically the main issue we are facing: at what point do kids need support? 30% of the sessions represented were early melodic interventions. The children's age groups of 6-8 and 9-12 as well as gaming sessions lasted for more than 3 hours were the situations with the highest risk. Thus, it is not simply the length of time the children are on their apps that counts, it is also the type of app and the time of the day they are using it. A child of 9 years who spends more than 30 minutes at night on social media is likely to require some kind of intervention much more than a 4-year-old who engages with a 10-minute educational application in the morning.

The Efficiency of the Prediction Model

A RandomForestClassifier was employed to study whether a session would be brought into the consideration for intervention, taking into account characteristics like session length, previous week's activity, app category, hour of the day, and age bracket. The model reached an accuracy of 89% on the test dataset and its ability to detect the “intervention required” cases was quite astonishing — it reached a precision of 0.87 and a very good F1-score. To sum up, the model indeed performs well in identifying sessions that are at risk.

Feature importance gave us a clear picture of what matters most. Here's what topped the list:

1. `duration_min`: The primary predictor that made the session the longest was along the line of a significant spike in their alertness level.
2. `frequency_last_7d`: Next was the use of the app frequently in the last week. Usage is habitual and habitual use in large volumes can trigger an alarm.
3. `app_category` (Game and Social): The risk of being in the Game or Social category really increased compared to the risk of using Education or Utility apps.
4. `time_of_day` (Evening): The evening was the time when intervention was most likely when using apps compared to any other time.
5. `age_group`: Older kids (6–8, 9–12) were caught a little more often, and could be that they are getting themselves in a cyber-complex social app.

In summary, the model not only alerts us to the risks, but also boundaries and allows us to monitor the situation before it escalates into a larger problem.

Discussion and Implications

Through this analysis, we can obtain a better understanding, supported by data, and manage kids' digital devices use more effectively. It is clear now that the screen time is not all the same. The app type, the duration of children's using it, the time of day, and their age are some of the factors that interplay to influence how kids feel and whether or not they need help.

The numbers relating to Social apps are really striking. Kids that use these applications are more likely to be identified as in need of intervention and the connection with being sad is very clear. That's a huge red flag for parents. Game apps are also a bit deceptive. In some cases, they are a source of happiness for children but in other cases, they can lead to long and frustrating sessions which alert the parents.

And this is the point in which the predictive model proves its worth. It collects all these revelations and makes them something that families can actually utilize. The most essential predictors, `duration_min` and `frequency_last_7d`, are now the basis for parents to keep track on specific numbers. The problem of "too much screen time" has been solved as a child's usage is simply a matter of the system's data-driven alarm for the specific period.

In the early childhood intervention system method, the hours of screen time are not the lone determining factor anymore. More considering the kids' actual online activities, new media time, and digital activities are still the kids' concern. Parents and teachers are provided with authentic, research-based recommendations that are helpful in managing kids' digital habits to be healthier and more balanced. This digital environment aims to be safe, educational and growth-promoting for young minds.

8. Limitations

To put it simply, the research has its flaws. The dataset has its restrictions, the model overlooks certain things, and there are still very important ethical dilemmas to be clarified before it can be deployed in the real world.

9. Conclusion

In the end, the project kept its word. An authentic data-driven system to detect children's risky online behavior was developed. The research indicated that it is not only the total screen time of children that is a problem but also their activities, the apps they use, and the timing of their usage. The foundation of the whole process is a predictive model that genuinely identifies problems at an early stage, thus giving adults the chance to step in before the situation gets worse. The households are now provided with easy-to-use tools and real data that can slowly lead their children from the beginning towards the adoption of more digital healthy habits.

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